

UNIT 1

Introduction

Sessions 1 and 2

This year the hardware is only half the work. You will be expected to specify before you build, measure your own error honestly, and defend your design decisions with evidence. That is the difference between making something work and engineering it.

By the end of this unit you will be able to:

- Write a requirement specification before starting a build.
- Name every component in your kit and say what it does.
- Distinguish accuracy from precision, and error from uncertainty.
- Explain why every measurement is a range rather than a single number.

SESSION 1

one class

Engineering a System, and What Is in Your Box

Today you will

- Learn what a requirement specification is and why it comes first.
- Learn the names of all sixteen components in your kit.
- Agree the rules of the robotics lab.

Anybody can wire a sensor to a buzzer. What separates an engineer is the order of the work. An engineer writes down what the machine must do **before** deciding how to do it, because otherwise there is no way to tell whether the finished thing is any good.

That written statement is a **requirement specification**. A good requirement is testable: somebody who did not build the machine must be able to read it, run a test, and say yes or no. Compare these two attempts at the same idea.

A poor requirement	Why it fails	A testable requirement
The alarm should be fast	How fast? Measured how?	The alarm shall sound within 500 ms of the beam being broken.
It should not give false alarms	Not measurable at all	The system shall produce no alarms during a 5 minute undisturbed test.
It should be accurate	Accurate to what?	Readings shall be within 1 cm of a tape measure between 10 and 150 cm.

Try This

Write three testable requirements for a school bell system: one about timing, one about volume, one about what happens during a power cut. Swap with another group and try to find a way each of their requirements could technically be met while still producing a useless bell. That exercise is how real specifications get tightened.

1.1 Open the Box

Lay everything out before connecting anything. The label colour on each tile tells you roughly which unit first needs that part. Two items deserve immediate respect: the laser module, which must never be pointed at anybody, and the battery pack, which must be disconnected whenever you rewire.



Figure 1.1 - Your kit. Learn these sixteen names before the next class.

Part	What it does	First used in
Arduino Uno board	The controller. It stores your program and runs it indefinitely.	Session 3
USB A-B cable	Carries the program in and powers the board during development.	Session 3
400-point breadboard	Solderless prototyping base.	Session 2
Jumper wires	Interconnections: male-male, male-female, female-female.	Session 2
5mm LEDs	Status indication: armed, alarm, ready.	Session 2
220 ohm and 10K resistors	Current limiting, and pull-downs for switch inputs.	Session 2
5V laser dot diode module	Emits the beam your alarm watches.	Session 6
LDR light sensor module	The receiver that catches the beam.	Session 6
16x2 LCD display	Shows the intrusion count and, later, live distance.	Session 7
10K potentiometer	Sets display contrast.	Session 7
Buzzer module	The alarm output.	Session 7
2-pin push switch	Clears the latched alarm.	Session 8
HC-SR04 ultrasonic sensor	The instrument you will characterise in Unit 4.	Session 9
IR sensor modules (three)	Read the maze: left, centre and right.	Session 12
L298N motor driver	Power stage between the controller and the motors.	Session 13

Part	What it does	First used in
Chassis with motors, wheels, caster and battery holder	The robot platform and its power source.	Session 13

1.2 Rules of the Robotics Lab

- 1 Always unplug the USB cable before changing any wire.
- 2 **Never aim the laser at anybody's eyes.** Mount it fixed, at bench level, pointing away from where people sit or walk.
- 3 Check polarity on every LED and module before applying power.
- 4 Never join 5V directly to GND. That is a short circuit.
- 5 Never connect a motor directly to an Arduino pin - the pin will be destroyed.
- 6 **Disconnect the battery pack before rewiring,** and at the end of every session.
- 7 Lift the robot off the floor before uploading, so it cannot drive away mid-upload.
- 8 Record every test in your engineering log, including the ones that failed.

Watch Out

Rule 2 is not negotiable. Even a low-power laser module is a collimated beam - all its energy arrives on a spot smaller than a pinhead, and your eye focuses it smaller still. The safe habit is simple: fix the module down before you power it, and keep the beam below eye level at all times.

SESSION 2

one class

Measurement, Error and Uncertainty

Today you will

- Distinguish accuracy from precision.
- Understand systematic error and random error.
- Express a measurement properly, as a value with an uncertainty.

Unit 4 of this book is an entire project about how wrong a sensor is. Before that can mean anything, you need the vocabulary - and most people, including many adults, use these words interchangeably when they mean completely different things.

Term	What it means	A sensor with the problem
Accuracy	How close a reading is to the true value	Reads 25 cm when the true distance is 30 cm
Precision	How closely repeated readings agree with each other	Reads 30, 34, 27, 33 cm for the same fixed object
Systematic error	A consistent offset or drift, the same every time	Always reads 3 cm too far - correctable
Random error	Unpredictable scatter around the true value	Wobbles a centimetre either way - reducible by averaging
Resolution	The smallest change the system can detect at all	Cannot tell 30.0 cm from 30.2 cm

The crucial insight is that accuracy and precision are independent. A sensor can be precise and wrong - giving you 25.0 cm every single time when the truth is 30. That is far more dangerous than a scattered sensor, because the consistency makes it look trustworthy. It is also, fortunately, the easier fault to fix: a systematic error can be measured once and corrected in software forever.

Behind the Scenes

This is why professional instruments carry a calibration certificate rather than a claim of perfection. The certificate does not say the instrument is right. It says how wrong it is allowed to be, and when it was last checked against a standard. Your Unit 4 project will produce exactly that kind of document for your own HC-SR04.

2.1 Writing a Measurement Properly

A single number is never a complete measurement. *The distance is 42 cm* hides the question a reader actually needs answered: how sure are you? The professional form is a value with an **uncertainty**: **42 plus or minus 2 cm**. That says the true value is very likely somewhere between 40 and 44.

You will estimate uncertainty the honest way: take several readings of the same fixed thing, and look at the spread. If ten readings of a wall at exactly 50 cm range from 48 to 52, your uncertainty is about 2 cm. No theory required - just data.

Try This

Measure the width of your bench ten times with a ruler, each person in the group taking a turn without looking at the previous answers. Write down all ten. Find the mean, the largest and the smallest. Now express the width as a value plus or minus an uncertainty. You have just performed the same procedure a national standards laboratory uses, only with cheaper equipment.

2.2 The Electrical Foundation

Everything in this book rests on one relationship, **Ohm's law**: $V = I \times R$. Voltage in volts equals current in amps multiplied by resistance in ohms. It is worth using it once properly rather than treating component values as magic.

Your LED drops about 2 volts across itself, so with a 5 volt supply the resistor must absorb the remaining 3 volts. With a 220 ohm resistor the current is 3 divided by 220, about 13.6 milliamps - comfortably within both the LED's rating and the 20 milliamp limit of an Arduino pin. Choose 22 ohms and you would demand 136 milliamps, and something would be destroyed.

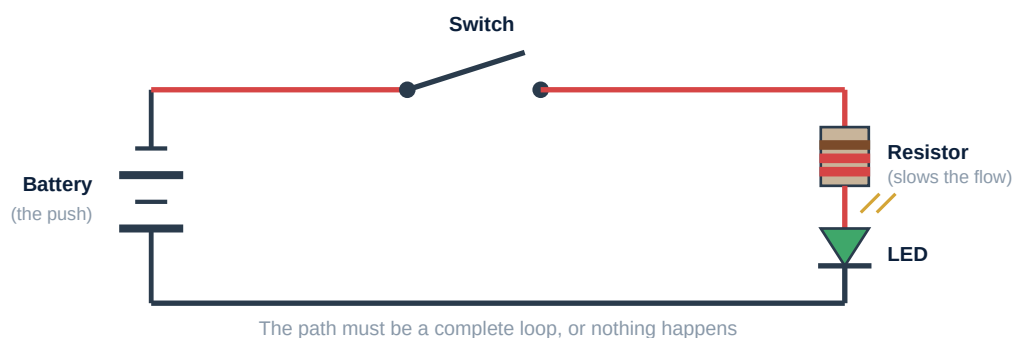


Figure 1.2 - A simple circuit: battery, switch, resistor and LED

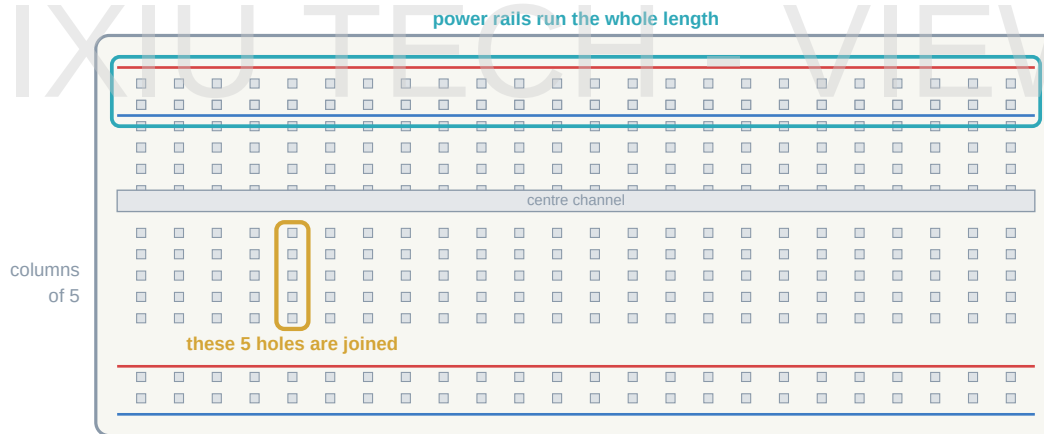


Figure 1.3 - Inside a breadboard: which holes are secretly connected

- The **long side rails** marked + and - run the full length. Use them for power and ground.
- Central holes are joined in **columns of five**, vertically only.
- The **centre channel** isolates the two halves, which is why chips straddle it.
- Two legs in the same column are connected; in different columns they are not.

New word	What it means
Requirement specification	A written, testable statement of what a system must do.
Accuracy	How close a reading is to the true value.
Precision	How closely repeated readings agree with each other.
Systematic error	A consistent offset or drift that can be corrected.
Random error	Unpredictable scatter, reducible by averaging.
Uncertainty	The range within which the true value very likely lies.